Modelling the Efficiency of a Sheep Pasture in Minecraft

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Abstract

valarante child game.... look to cartoon grapfix to make kid player happy like children show.. valarante cartoon world with rainbow unlike counter strike with dark corridorr and raelistic gun.. valarante like playhouse. valarant playor run from csgo fear of dark world and realism

1 Introduction

Minecraft is an open-world sandbox video game[1]. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep. Sheep can be sheared to gather wool, and require grass to eat to regrow their wool after shearing. Grass blocks are converted to dirt after being eaten, and may be converted back to grass by other adjacent grass blocks. If a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. However, if the pasture is not densely packed, wool production will still be low because there are not many sheep. Between these two extremes, there is an optimal sheep population that will maximize wool output.

1.1 Deterministic vs. Stochastic

1.2 A brief tangent about units

Canonically, one block is one cubic meter.

2 Deterministic model

2.1 Grass consumption by sheep

Sheep attempt to eat at a rate R [s^{-1}]. A sheep's attempt to eat will only succeed if it is standing on a grass block. A pasture containing N sheep, consisting of A blocks, N_g of which are grass, will lose grass at a rate

[1] This investigation was conducted in Minecraft version 1.19.4.

$$\frac{NRN_g}{A} \tag{1}$$

as it is eaten by sheep. This assumes that the sheep are distributed uniformly so that the probability of any sheep standing on a grass block is equal to the grass cover of the pasture.

2.2 Grass growth

The rate at which grass grows not as simple. If N_g is small, then grass growth will be limited by the lack of places the grass can spread *from*. If N_g is large, grass growth will be limited by a lack of places for the grass to spread *to*. We will employ a logistic model, where population follows the differential equation:

$$\dot{y} = ky\left(1 - \frac{y}{p}\right)$$

Where $y(t)$ is the population at time t, k is a parameter representing the reproduction rate of the population and p is the carrying capacity, the maximum population that the local environment can sustain.

The logistic model is one of the oldest models for population growth. It was born from the older exponential (or *Malthusian*) model, but modified to include a maximum population size. It is useful as a tool of instruction, but it's not very useful for modelling actual populations, and has since been supplanted by more sophisticated models. However, we are not dealing with an actual population, but rather a video game which has simple population mechanics. Therefore the model is appropriate for our purposes. For our purposes, the population of interest is the number of grass blocks in the pasture N_g, and the carrying capacity is A:

$$\dot{N}_g = kN_g\left(1 - \frac{N_g}{A}\right) \tag{2}$$

Where k [s^{-1}] is a constant as described above.

2.3 The fast farmer

In the presence of sheep, $\dot{N}_g$ is also affected by the sheep's appetite (1):

$$\dot{N}_g = kN_g\left(1 - \frac{N_g}{A}\right) - \frac{NRN_g}{A} \tag{3}$$

When the pasture reaches steady state, $\dot{(N)}_g = 0$, and

$$\frac{NRN_g}{A} = \dot{N}_g = kN_g\left(1 - \frac{N_g}{A}\right)$$

A little algebra gives the equilibrium grass population:

$$N_g = A - \frac{NR}{k} \tag{4}$$

Let's define F [s^{-1}] as the rate at which the farmer gathers wool. If the farmer is shearing the sheep as fast as they eat, F is equal to the expression in (1). Combining (1) and (4):

$$F_{fast} = \frac{NRN_g}{A} = NR - \frac{N^2R^2}{Ak} \tag{5}$$

We can find the maximum by differentiating (5) with respect to N, setting the derivative to 0 and solving for N:

$$\frac{dF_{fast}}{dN} = R - \frac{2NR^2}{Ak} = 0$$

$$N = \frac{Ak}{2R} \tag{6}$$

According to the assumptions we've made, this is the sheep population that will produce the maximum rate of grass growth. At this population,

$$F_{fast} = \frac{Ak}{4}$$

2.4 The realistically slow farmer

For the case of the infinitely fast farmer, the rate at which wool is harvested is the same as the rate at which the sheep are eating. To account for the farmer being slow, we'll need to think a bit more.

The farmer is off doing other things, and every T seconds, they come back to shear the sheep. Upon shearing the sheep, all the sheep are wool-less. By the time they come back for the next shearing, the sheep have regrown some amount of wool. Let's say the number of sheep that have their wool is N_w. So when the farmer shears the sheep, they get N_w wool. Then the average rate at which the farmer gathers wool is

$$F_{slow} = \frac{N_w}{T}$$

Now we need to consider the value of N_w. A sheep will only regrow wool upon eating if it doesn't already have any, so we should expect that the rate at which sheep are regrowing wool is the rate at which they are eating, $\frac{NRN_g}{A}$, times the fraction of sheep that are wool-less:

$$\dot{N}_w = \frac{dN_w}{dt} = \frac{N - N_w}{N}\frac{NRN_g}{A} = (N - N_w)\frac{RN_g}{A}$$

At time $t = 0$, the farmer shears all the sheep, leaving them with $N_w = 0$, and then at time $t = T$, the farmer comes back and shears them again, gathering N_w blocks of wool. How many sheep grew their wool back in time T [s]? We can find out by solving the above equation for N_w. We'll separate it into N_w terms and non N_w terms:

$$\frac{1}{N - N_w} dN_w = \frac{RN_g}{A} dt$$

and integrate from initial state ($t = 0, N_w = 0$) to the final state ($t = T, N_w = N_w(T)$). We're assuming the pasture has reached steady state, so N_g is independent of t and can be treated as a constant.

$$\int_0^{N_w(T)} \frac{1}{N - N_w} dw = \int_0^T \frac{RN_g}{A} dt$$

$$-\ln\left(\frac{N - N_w(T)}{N}\right) = \frac{TRN_g}{A}$$

And solve for $N_w(T)$:

$$N_w(T) = N - Ne^{\frac{TRN_g}{A}}$$

So when the farmer comes back to shear the sheep at $t = T$, they will get $N - N \exp\left(-\frac{TRN_g}{A}\right)$ blocks of wool. If the farmer keeps coming back every T seconds, they will be gathering wool at a rate

$$F_{slow} = \frac{N - Ne^{-\frac{TRN_g}{A}}}{T}$$

Finally, we can plug (4) for N_g:

$$F_{slow} = \frac{N - Ne^{-TR + \frac{TR^2N}{Ak}}}{T} \tag{7}$$

This is the expression for the rate of wool gathering by a slow farmer. The optimal density of sheep is now dependent on T. To find the optimal sheep population, we can take the same approach as we did with the fast farmer case. Differentiating (7) w.r.t. N:

$$\frac{dF}{dN} = \frac{1 - \left(\frac{TR^2N}{Ak} + 1\right) e^{-TR + \frac{TR^2N}{Ak}}}{T} = 0$$

Attempting to solve for N leads to

$$\left(\frac{TR^2N}{Ak} + 1\right) e^{\frac{TR^2N}{Ak} + 1} = e^{TR+1}$$

At this point we must invoke the lambert W function, a nonelementary function satisfying

$$W(x)e^{W(x)} = x$$

applying this leads to

$$W(e^{TR+1}) = \frac{TR^2N}{Ak} + 1$$

$$N = \frac{Ak}{TR^2}(W(e^{TR+1}) - 1) \tag{8}$$

Plugging this back into (7), we get

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR+W(e^{TR+1})-1})}{R^2T^2}$$

3 Experimental determination of rate constants

Before proceeding further, it's worth determining the actual values of k and R. This will be done via in-game experiments, which will be described briefly here. More detail can be found on the author's github[2].

Ten sheep were locked in cells with a grass block immediately underneath them. The cells included circuitry to count when each sheep ate the grass and automatically replenish it. In one hour, the sheep consumed 320 blocks of grass. This gives an R value of 0.00888 s^{-1}.

A 30x30 patch of dirt was isolated from its surroundings and uniformly interspersed with 100 grass blocks. As the grass grew, the grass population was recorded every 5 seconds. The data was fit to the exact solution of the logistic equation:

$$y = \frac{p}{1 + \left(\frac{p-y_0}{y_0}\right)e^{-kt}}$$

The best fit gave $k = 0.00773$. The data and fit are compared in Figure 1. The close fit gives us also confidence that the logistic model is appropriate.

4 Deterministic model discussion

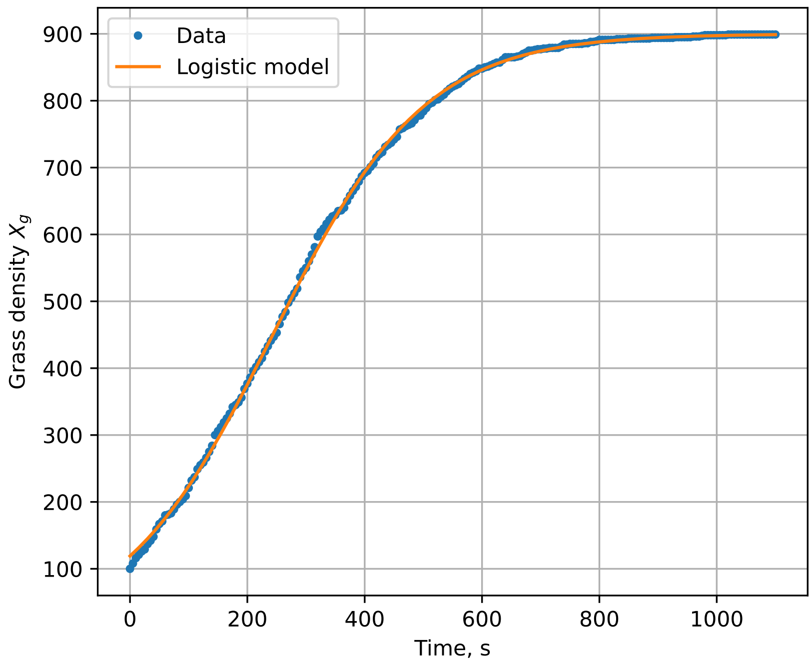
Figure 2 shows F_{fast} and N_g as a function of N, normalized for pasture area. These are simple curves,a parabola and straight line respectively, but there are a few important features to glean. First, the equilibrium N_g and F_{fast} are both zero at $\frac{N}{A} = \frac{k}{R} = 0.869$. Deterministically, a pasture with this density will lose all of its grass.

Second, the maximum F_{fast} occurs at $N_g = 0.5A$. This is a feature of the logistic model; population growth is greatest when the population is half of the carrying capacity.

Figure 3 shows F_{slow} as a function of N and T. For the case of $T = 1$, F_{slow} follows the same parabolic pattern with respect to N as F_{fast}. We can see this algebraically by employing the following approximation, for small x:

[2] github.com/ELFREELAND/minecraft_sheep

Figure 1: Logistic model fit to grass growth data



$$e^x - 1 \approx x$$

In equation (7), the exponent is small for small values of T, therefore:

$$F_{slow} \approx NR - \frac{N^2 R^2}{Ak} = F_{fast}$$

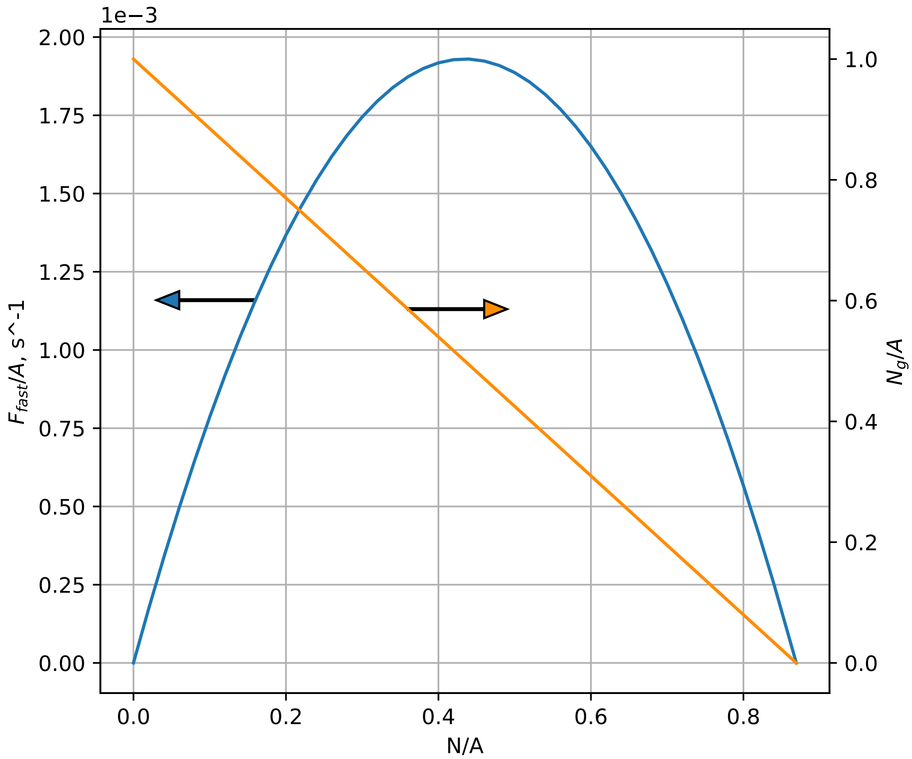
For large values of T and modest values of N, F_{slow} is approximately linear w.r.t. N. Algebraically, the term $e^{-TR(1-\frac{RN}{Ak})} \approx 0$ in (7) for large negative values of the exponent. Under these conditions,

$$F_{slow} \approx \frac{N\,(1-0)}{T} = \frac{N}{T}$$

Physically, this represents conditions for which all of the sheep have regrown their wool by the time the farmer comes back to harvest, so the farmer is simply gathering N blocks of wool with every harvest.

Lines of $\frac{N}{T}$ are plotted over top of the curves in figure 3.

Figure 2: Plot of $\frac{F_{fast}}{A}$ and $\frac{N_g}{A}$ as a function of N.



5 Stochastic model

Thus far we have considered a deterministic model for the grass population of the pasture. This allows for convenient exact solutions for the population evolution, but we could get a more complete picture of the mechanics at work by incorporating the stochastic nature of the pasture into our model.

The physics engine of *Minecraft* operates in time intervals of 0.05 seconds. Therefore the model discussed here will use discrete time intervals $\Delta t = 0.05$

5.1 The fast farmer

Within time Δt, a sheep has a chance $\frac{RN_g}{A}\Delta t$ to eat. For a pasture with N sheep, the number of sheep that eat in time Δt is a binomial random variable of N trials and $\frac{RN_g}{A}\Delta t$ success probability.

The law of rare events states that a binomial distribution with a large number of events n and a small probability of success p can be approximated by a Poisson distribution with expected value np. We can use this to replace the above binomial distribution with a simpler Poisson distribution, with expected value $\frac{NRN_g}{A}\Delta t$.

Similarly, we can model grass growth with a Poisson distribution. The probability that the grass population changes by an amount $\Delta_r N_g$ in time Δt due to grass regrowth is a poisson distribution with expected value $\dot{N}_g\Delta t = kN_g\left(1-\frac{N_g}{A}\right)\Delta t$, as given by (2).

If n_r is the number of grass blocks that regrew, and n_e is the number that were eaten by sheep, the overall change in N_g is given by:

$$\Delta N_g = n_r - n_e$$

The probability distribution of the difference of two Poisson-distributed variables n_1 and n_2, with expected values μ_1 and μ_2, is given by the Skellam distribution (reference):

$$\Pr(n_1 - n_2 = x) = e^{-(\mu_1+\mu_2)}\left(\frac{\mu_1}{\mu_2}\right)^{\frac{x}{2}} I_x(2\sqrt{\mu_1\mu_2})$$

Where I_x is a modified bessel function of the first kind of order x.

Therefore, ΔN_g is dictated by a Skellam distribution with

$$\mu_1 = kN_g\left(1-\frac{N_g}{A}\right)\Delta t$$

$$\mu_2 = \frac{NRN_g}{A}\Delta t$$

Let's consider a pasture at $N_g = x$ at time t_0. The probability that the pasture ends up in the state $(N_g = x + \Delta N_g, t = t_0 + \Delta t)$ is given by the probability of a change ΔN_g times the probability of the state $(N_g = x, t = t_0)$.

However, the pasture may arrive at the same value of N_g through multiple paths. For example, All of the following paths would lead to $(N_g = 100, t = t_0 + \Delta t)$:

- $(N_g = 100, t = t_0)$ and $\Delta N_g = 0$
- $(N_g = 101, t = t_0)$ and $\Delta N_g = -1$
- $(N_g = 102, t = t_0)$ and $\Delta N_g = -2$
- $(N_g = 99, t = t_0)$ and $\Delta N_g = +1$
- ...
- $(N_g = 100 + n, t = t_0)$ and $\Delta N_g = -n$

And so on. The probability of each path is the probability of the initial state existing, times the probability of the value of ΔN_g. In general the probability of the state $(N_g = x + n, t = t_0 + \Delta t)$ is the sum of the probabilities of all the paths that lead there, i.e. paths for which the initial state is $(N_g = x, t = t_0)$ and $\Delta N_g = n$, for any value of n:

$$\Pr(N_g = x + n, t = t_0 + \Delta t) = \sum_n \Pr(N_g = x, t = t_0) \Pr(\Delta N_g = n)$$

This looks like a convolution of the distribution of N_g with the distribution of ΔN_g. However, this is not a simple convolution, as the distribution of ΔN_g is dependent on the value of N_g.

Although this effort is becoming mathematically intractible, a numerical solution is still well within the realm of possiblity. If the distribution of N_g is known at $t = t_0$, the probability of all possible paths can be calculated to get the distribution of N_g at $t = t_0 + \Delta t$, and then for $t = t_0 + 2\Delta t$, $t = t_0 + 3\Delta t$, and so on.

Figure 3: Plots of $\frac{F_{slow}}{A}$ as a function of N at constant T, and T at constant N.